VoiceOps 2026: Multilingual Support Agent

LLM & Generative AI Course Capstone · Consulting & Analytics Club, IIT Guwahati × Gnani.ai

1. Executive Summary & Architecture

VoiceOps is an automated customer support solution built for **ZentroMart**, an Indian online store handling groceries, electronics, and fashion. The architecture unifies speech understanding (Gnani STT), context-gated conversational reasoning, hybrid RAG over 21 company policies (82 sections), offline identity verification, and human escalation with voice response generation (Gnani TTS).

Component	Implementation & Role	Policy / Spec
Speech Layer	Gnani STT (REST) & TTS (timbre-v2.5, Kaveri voice)	Step 4 / docs.gnani.ai
RAG Engine	Hybrid TF-IDF/Vector + Entity Linking over 82 KB sections	Step 2 / KB-00.1
Verification	Caller identity gate: phone last 4 digits OR pincode	Step 3 / IDV-1..IDV-4
Calendar SLA	Mon-Sat business days math, skipping holiday calendar	KB-00.3 / 20 Oct 2026
Escalation	Auto-ticket generation for damage > ■10k or repeat calls	ESC-1 / ESC-2

2. Engineering the Four Steps

Step 1: Baseline Prompting Assistant

Configured a baseline conversational LLM instructed to introduce itself per [STY-1]: *'I am ZentroMart's AI support assistant.'* Without access to company documents or tools, the baseline relies solely on pre-trained parametric knowledge, failing domain inquiries.

Step 2: Grounded RAG with Precedence Arbitration

- **Section Chunking:** Chunks 21 Markdown files in kb/, preserving section IDs (e.g. [RET-2]) and frontmatter metadata.
- **Document Precedence [KB-00.1]:** Current policies (1.3x) > Agent guides (1.15x) > Help (1.0x) > FAQs (0.85x) > Marketing (0.65x) >> Superseded (0.2x).
- **Catalog Join:** Automatically matches SKUs (e.g. ELE-0012) to products.csv for warranty and return categories.
- **Honest Fallbacks:** When information is missing (e.g. midnight deliveries), the agent states lack of coverage rather than hallucinating.

Engineering the Four Steps (Continued)

Step 3: Multi-Turn Conversation, Identity Gate & Tools

- **Identity Verification Gatekeeper:** Strictly gates order details behind lookup_order(order_id). Details are revealed only after verifying the registered phone's last 4 digits or delivery pincode. Max 1 retry is enforced per [IDV-3].
- **Date Engine (Reference: 20 Oct 2026):** Accurately differentiates calendar-day return windows (e.g. 15 days for fashion, 7 days for electronics) from business-day refund SLAs (Mon-Sat, skipping warehouse holidays such as 19 Oct 2026).
- **Human Escalation [ESC-1]:** Automatic ticket generation via create_ticket() when damaged item exceeds ■10,000, repeat contacts ≥ 2, refunds exceed SLA by > 3 days, or customer asks for a human.

Step 4: Multilingual Voice Layer (Gnani STT & TTS)

- **Speech Pipeline:** Integrates official gnani-vachana SDK (STT transcribe & TTS synthesize with Kaveri voice, timbre-v2.5).
- **Writing for the Ear [STY-3]:** Normalizes order IDs into grouped digits with pauses ('8 9 7 8, 2 8 3 7'), formats amounts as spoken words ('forty-nine rupees'), and speaks dates with full month and year ('the 24th of October 2026').
- **Silence & Error Handling:** In case of silent or distorted audio, gently reprompts: *'Sorry, I didn't catch that. Could you say it again?'*

3. Empirical Evaluation & Quantitative Results

The system was evaluated against the official 52 practice questions (including simulated ASR speech transcriptions) and 23 multi-turn conversations (96 total customer turns).

Metric	Step 1 Baseline	Step 2 Grounded RAG	Improvement
Fact Groundedness Accuracy	5.8%	**86.5%**	**+80.7%**
Evidence Retrieval Recall	0.0%	**74.7%**	+74.7%
Evidence Retrieval Precision	0.0%	**38.0%**	+38.0%
Evidence Retrieval Hit Rate	0.0%	**86.5%**	+86.5%
Hallucination Rate	94.2%	**0.0%**	**-94.2%**

Step 3 Multi-Turn Dialogue Benchmark (96 Customer Turns)

Evaluation Criterion	Expected Target	VoiceOps Measured Score
Conversation Decision Match (clarify/answer/escalate/refuse)	≥ 90.0%	**97.9%** (94 / 96 turns)
Tool Invocation Accuracy (lookup_order, create_ticket)	≥ 90.0%	**97.9%** (53 / 54 tool calls)
Customer Data Verification Leakage Rate	0.0%	**0.0%** (Zero private leaks)
Multi-turn Evidence Recall	≥ 80.0%	**86.8%**

4. In-Depth Error Analysis & Technical Lessons

A rigorous audit of edge-case failures during iterative development revealed three primary operational failure modes:

A. Speech-to-Text Phonetic Distortions & ASR Number Homophones

• *Symptoms:* Simulated ASR inputs rendered SKUs phonetically (e.g. ele-zero zero one to instead of ELE-0012) and years as words (two zero two six instead of 2026). Standard regex parsers failed to extract product entities.
• *Resolution:* Engineered a phonetic query preprocessor (normalize_asr_query) that detects phonetic digit sequences, maps acoustic homophones (to / too $\rightarrow$ 2, for $\rightarrow$ 4), collapses fragmented acronyms (u p i $\rightarrow$ upi, re fund $\rightarrow$ refund), and links product brand names to SKUs.

B. Superseded Policy Contamination & Version Collisions

• *Symptoms:* When answering electronics return window questions, the retriever surfaced both pol_returns_v4.md (7 days) and pol_returns_v3_superseded.md (10 days). LLMs frequently averaged or selected the superseded document.
• *Resolution:* Implemented precedence scoring in PrecedenceRanker, severely demoting status: superseded chunks (0.2x multiplier) unless the query specifically referenced historical delivery dates (e.g., delivered before 1 August 2026).

C. Non-Standard Warehouse Festival Holidays

• *Symptoms:* Refund SLA calculations for credit cards (5-7 business days) and UPI (1-3 business days) initiated on Saturday 17 October 2026 miscalculated arrival dates when standard Monday-Saturday counting skipped only Sunday.
• *Resolution:* Embedded data/holidays.csv into date_utils, properly skipping Monday 19 October 2026 (festival closure) and yielding the exact expected deadline (27 October 2026 for credit cards, 22 October 2026 for UPI).

5. Production Roadmap & What We Would Fix

While VoiceOps exceeds benchmark criteria, the following enhancements would be deployed for full-scale commercial production:

1. **Streaming Full-Duplex Voice over WebSocket:** Transition from REST-based batch audio to bidirectional WebSocket audio streaming using Gnani's real-time streaming endpoint, reducing perceived latency to under 400ms.
2. **Two-Stage Cross-Encoder Reranking:** Integrate a lightweight cross-encoder (e.g. BGE-reranker) to refine top-10 hybrid search candidates, improving fine-grained contextual precision.
3. **Multi-Factor Voice Biometrics:** Augment 4-digit phone verification with caller voice biometrics to prevent SIM-swap fraud and enhance caller security.

6. Bonus Work & Full Portfolio Assets

To ensure VoiceOps serves as an exemplary open-source AI engineering portfolio project, four major bonus systems were implemented:

Bonus Feature	Technical Capabilities
Streamlit Web Application (app.py)	Full-featured browser UI supporting live microphone audio recording, interactive multi-turn chat, real-time tool state visualization, RAG policy explorer, and benchmark metric charts.
Automated CLI Runner (run.py / run.sh)	Fulfills the official submission testing contract: processes any arbitrary directory of .wav files, transcribes speech, invokes RAG agent brain, synthesizes speech, and outputs structured JSON citations.
Automated Pytest Suite (tests/)	13 unit and integration tests verifying date calculations, business day logic, identity verification barriers, tool execution, and voice text normalization with 100% pass rate.
Offline Fallback Engine (Zero-Crash Guarantee)	The entire test pipeline, batch runner, and web UI can run offline with zero external API dependencies or seamlessly connect to live Google Gemini and Gnani.ai APIs when keys are configured.

7. Verification of Submission Deliverables

All required deliverables from Capstone Brief Page 14 are verified and present in the codebase:

Deliverable	Location	Verification Status
Code Repository & README	README.md, src/	■ Complete & Documented
Environment Template	.env.example	■ No keys committed
Batch Run Command	python run.py --input_dir ...	■ Validated on audio batch
Technical Report	report.pdf, REPORT.md	■ Exact 4-page format
Evaluation Pipeline	eval_pipeline.py	■ Step 1 vs Step 2 vs Step 3
Sample Outputs	sample_outputs/batch_run/	■ Text, audio & citations

Report compiled automatically for IIT Guwahati Consulting & Analytics Club × Gnani.ai Capstone 2026.